
Danial Younas

SYSTEMS & DATA ENGINEER

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Summary

Systems & data engineer with 2+ years of production experience designing and operating cloud-native infrastructure under live operational pressure. Sole engineer behind a smart-city telemetry platform serving 3,000+ devices across seven zones, integrating multi-vendor field systems into one real-time data layer using Go, AWS, Kafka, PostgreSQL, Docker, MQTT, Modbus, and LoRaWAN. Strong at solving difficult integration problems, building reliable ingestion systems, and keeping critical infrastructure running without interruption.

Experience

Systems & Data Engineer

Smart TechOne • Islamabad, Pakistan

01/2024 – Present

Sole engineer responsible for architecting, deploying, and operating the cloud-hosted data acquisition platform supporting Capital Smart City. Built the stack end to end from field ingestion through observability and dashboard delivery, replacing fragmented vendor-locked systems with a unified real-time platform.

- Architected and deployed a Go-based multi-service data acquisition platform on AWS, normalising telemetry from 3,000+ devices across seven operational zones into a unified cloud-native pipeline.
- Integrated multi-vendor devices across Energy Management, Air Quality, Water Management, and Waste Management using Modbus RTU/TCP, LoRaWAN, MQTT, and custom vendor communication protocols.
- Containerised platform services with Docker and currently leading migration from on-premises Linux servers to AWS-hosted container infrastructure with zero-downtime cutover planning.
- Implemented CI/CD pipelines using GitHub Actions and Jenkins for automated builds, service packaging, and controlled environment deployments.
- Deployed and operate Grafana and Prometheus observability for 50–100 active data streams, with custom alerting for ingestion health, SLA breaches, and field communication failures.
- Designed and operate a WireGuard VPN mesh securing 50+ distributed gateways across city zones, including routing, firewall rules, DNS/DHCP, and remote access controls.
- Reduced new device onboarding from 3–4 days to a matter of hours through scripted provisioning, reusable integration patterns, and payload-decoding automation.
- Implemented buffering and batching in the ingestion layer, reducing server resource consumption by 40% while maintaining real-time SLAs.

Selected Platform & Domain Highlights

- Energy, air-quality, and water systems integrated via Modbus RTU/TCP with vendor-specific register maps and custom parsing logic to surface operational metrics.
- Waste sensors and smart water meters integrated via LoRaWAN (Loriot LNS), with proprietary binary frames decoded and normalised into a unified *DeviceReading* JSON model.
- Event streams routed through Apache Kafka into PostgreSQL on AWS, feeding live dashboards and the structured data layer used by the city's digital twin platform.

Self-Learning Projects

GitOps Pipeline Lab

Argo CD · Kubernetes · GitHub Actions

- Built a local GitOps deployment workflow where Argo CD syncs application manifests from GitHub to a Kubernetes cluster.
- Practised progressive delivery patterns including blue-green and canary-style rollout strategies.

HPC Cluster & SLURM Workload Manager

SLURM · PyTorch · Prometheus · Grafana

- Built a SLURM-based HPC cluster on Fedora with shared NFS storage and batch-submitted GPU-accelerated PyTorch training jobs.
- Integrated Prometheus and Grafana to monitor queue depth, node health, and workload utilisation.

Terraform on AWS

Terraform · AWS · IAM · VPC

- Provisioned reusable AWS infrastructure for networking, compute, security groups, and IAM using modular Terraform configurations.
- Practised repeatable environment setup and remote-state workflows for infrastructure changes.

Linux & Jenkins Lab

Linux · Jenkins · Bash · CI/CD

- Built a Linux-based automation lab for Jenkins pipelines, service deployment practice, user/access administration, and repeatable build workflows.
- Used shell scripting to automate common operational tasks and reinforce production-style CI/CD habits.

Skills

Languages: Go, Python, SQL, Bash

Cloud & Platform: AWS, Azure, Docker, Kubernetes, Linux, Terraform, Ansible

Data & Observability: PostgreSQL, Apache Kafka, MQTT, Grafana, Prometheus

IoT & Networking: LoRaWAN, Modbus RTU/TCP, WireGuard, TCP/IP, DNS, DHCP

Delivery: GitHub Actions, Jenkins, CI/CD

Education

Bachelor's Degree in Computer Science

Virtual University of Pakistan · Islamabad, Pakistan

Graduated 10/2024